

Building a Simple Application using Streamlit on Python

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Abstract—Streamlit is an open source library designed to easily develop data-driven and machine learning web applications using Python. Launched in 2018, Streamlit allows developers to quickly turn data scripts into interactive web applications without writing backend code, configuring routes, or handling HTTP requests. In this report, we will learn a basic introduction to the Streamlit framework and its implementation to build a simple application. The use of Streamlit proves that Streamlit allows developers to create web applications without needing to master HTML, CSS, or JavaScript. You just need to write Python code.

Keywords—Streamlit, Python, Data Science, Machine Learning, Web Application

I. INTRODUCTION

Python is one of the most popular programming languages due to its ease of syntax and flexibility in various domains, such as data analysis, web development, and artificial intelligence.[1] Web-based application development is increasingly important in today's digital era to meet various needs, from business to education. One tool that supports the development process is Visual Studio Code (VS Code), a code editor that is popular because of its reliability, user-friendly interface, and extensive extension support. VS Code also allows developers to easily create simple applications by integrating various Python libraries, such as Streamlit. Streamlit is a Python framework intended to make developing interactive web applications easier. With this framework, users can build application interfaces without having to understand front-end technologies such as HTML, CSS, or JavaScript.[2] Streamlit is a great choice for professional developers and beginners alike because of its automatic rendering, user input management, and data visualization.

The goal of this project is to create a simple application that uses VS Code and Streamlit. The development process includes installing tools, writing code, and running the application locally. These applications can do basic things like accept user input, process data, and display interactive visualizations. It is hoped that this project will teach students or novice developers the first steps in using contemporary technology to create simple web applications.

II. BASIC STREAMLIT FRAMEWORK

First, readers should learn the basics of the streamlit framework before starting to build simple applications using streamlit. The basis of the streamlit framework consists of

simple setup, basic commands, interactivity with widgets, including displaying data, session state, and deployment.

1. Simple setup

First, make sure the reader has to install streamlit first via the command prompt using the pip install streamlit command.

```
Command Prompt
Microsoft Windows [Version 10.0.22621.3810]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>pip install streamlit
```

Then, readers can confirm whether it has been installed by using the streamlit hello command at the command prompt.

```
C:\Users\USER>streamlit hello

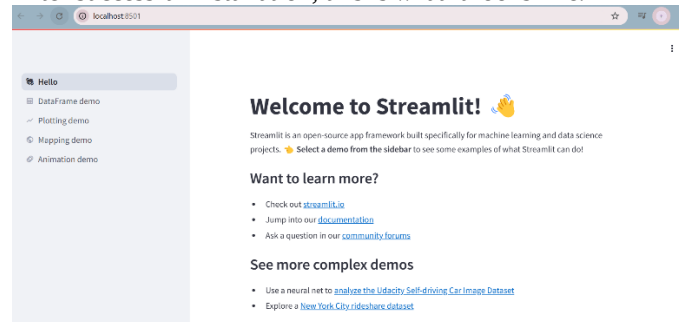
Welcome to Streamlit. Check out our demo in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.1.87:8501

Ready to create your own Python apps super quickly?
Head over to https://docs.streamlit.io

May you create awesome apps!
```

After successful installation, this is what it looks like.



2. Basic commands

Streamlit offers a number of built-in features that allow creating various parts of the user interface:

- St. title: Add a main title at the top of the page.
- St. header: add headers or sub-headers.
- St. text: displays text.
- St. write: add plain text.

3. Interactivity with widgets

Many of the interactive widgets that Streamlit supports allow users to interact with applications and import data:

- St. slider: slider for number selection.
- St. button: button to trigger an action when clicked.
- St. selectbox: Allows users to select options from a dropdown.[3][4]

4. Display data

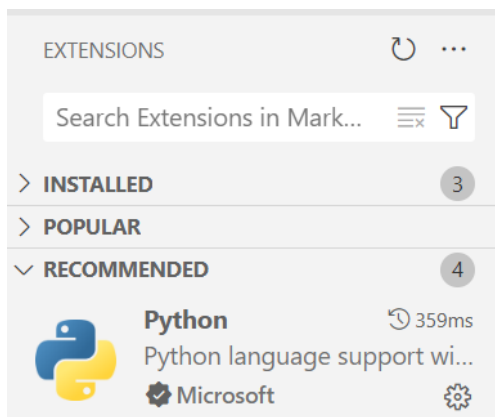
Using functions like `st.dataframe` for tables and `st.line_chart` for graphs, you can display data easily. The `st.write` function is very useful because it can handle various types of data quickly.

5. Session status

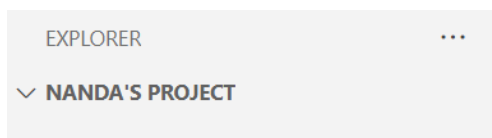
Especially useful for applications that require persistent user data because Streamlit maintains session state and allows you to track user input during interactions. You can easily share your app when you're done using Streamlit Community Cloud or other cloud platforms.

III. BUILDING THE WEB APPLICATION

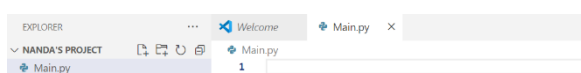
After learning about basic streamlit commands, readers will then start the stages of building a simple web application. The first step that must be done is to install VS Code. After it has been successfully installed, the next step is to install Python on VS Code.[5] The results are shown in the image below.



After Python is successfully installed, the next step is to create a new folder in VS Code.



Then, right click select new file and give the file a name.



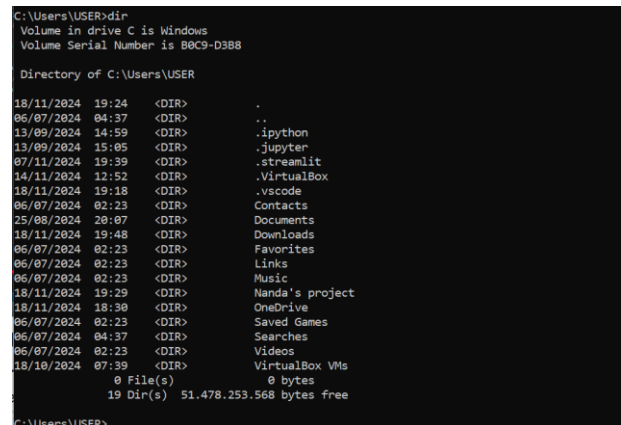
Next, enter the code to create a simple web application.

```

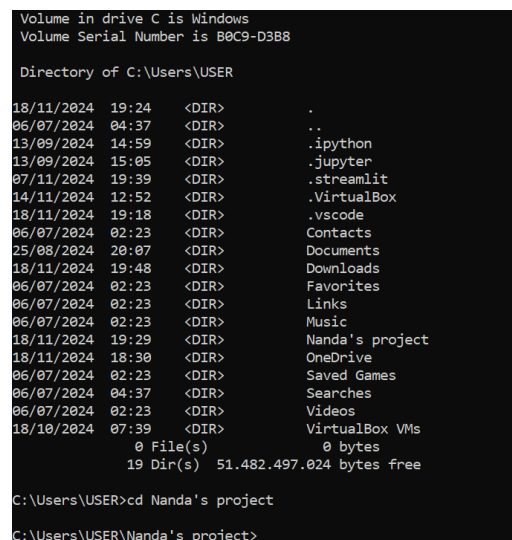
Main.py > ...
1  import streamlit as st
2
3  st.title('Hitung Luas Persegi Panjang')
4
5  panjang = st.number_input("Masukkan Nilai Panjang", 0)
6  lebar = st.number_input("Masukkan Nilai Lebar", 0)
7  hitung = st.button("Hitung Luas")
8
9  if hitung:
10     luas = panjang * lebar
11     st.write("Luas Persegi Panjang Adalah = ", luas)
12     st.success(f"Luas Persegi Panjang Adalah = {luas}")

```

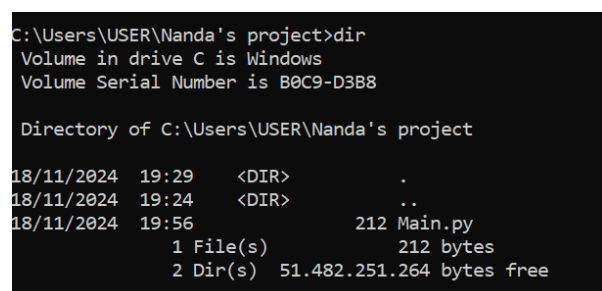
To run the streamlit application, we can enter the command prompt, then type `dir` to see the contents of a directory where we can save the files we created.



Click `cd` to enter Nanda's project directory.



Next, click `dir` to see Nanda's project directory.



Then, run the application using the streamlit run Main.py command.

```
C:\Users\USER\Nanda's project>streamlit run Main.py
```

Streamlit will provide a local link, such as <http://localhost:8501>, that can be opened in a browser to view your application.

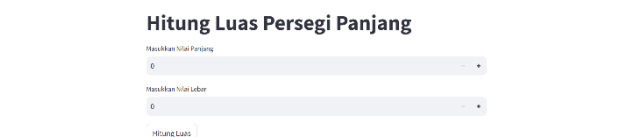


That's how to create a simple web application using streamlit. Readers must follow the steps given. Starting with the installation of Python, Streamlit, and VS Code, setting up the development environment is very important. Then, the procedure continues with creating a Python file in VS Code to write application code that uses basic Streamlit components, such as input, output, and visualization. Finally, the Streamlit command allows applications to be run through the terminal, allowing developers to view and test them directly in the browser.

IV. RESULT AND DISCUSSION

This is the result of creating a simple web application regarding how to calculate the area of a rectangle in the previous chapter after using streamlit to build the application.

1. Starting from the rectangular area calculation display



2. Enter the number that the user wants to calculate.

V. CONCLUSION

Developers, especially those who want to avoid the complexity of front-end technology, can use Streamlit to build simple web applications quickly, easily and effectively. This framework is very suitable because it allows creating interactive user interfaces with just Python. With easy-to-use features, interactive components, and Library integration, Streamlit makes creating and converting Python code into accessible web applications a very easy task.

VI. ACKNOWLEDGMENT

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- [5] "Visual Studio Code Documentation," Microsoft Corporation. Accessed: Nov. 25, 2024. [Online]. Available: <https://code.visualstudio.com/docs>

STATEMENT

I agree with this statement that my writing is not plagiarism or a translation of other people's work.

Semarang, 19 November 2024

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